MCS-102	FOUNDATION OF COMPUTER SCIENCE	L T P 3 1 0	4 Credits

Course Objectives:

CO1	An ability to apply knowledge of computing and mathematics appropriate to the discipline
CO2	An ability to identify, formulate, and develop solutions to computational challenges.
CO3	An understanding of professional, ethical, legal, security, and social issues and responsibilities for the computing profession.
CO4	Recognition of the need for and ability to engage in continuing professional development
CO5	An ability to apply design and development principles in the construction of software systems of varying complexity.

MODULE I

Regular languages

Sets, functions, Relation, Alphabet, Languages and grammars. Regular grammars, regular expressions and finite automata, deterministic and nondeterministic. Closure and decision properties of regular sets. Pumping lemma of regular sets. Minimization of finite automata.

MODULE II

Context free Languages

Context free grammars and pushdown automata. Chomsky and Griebach normal forms. Cook, younger and Kasami Algorithm, Ambiguity and properties of context free languages pumping lemma. Deterministic pushdown automata. Closure properties of deterministic context free languages.

MODULE III

Turing Machine

Turing machines and variation of Turing machine model, Halting problem, Universal turing machine, Type 0 Languages. Linear bounded automata and context sensitive languages. Turing Computable functions, Church Turing hypothesis. Recursive and recursively enumerable sets, Universal Turing machine and undecidable problems, Rice's theorems for RE sets, Undecidablity of Post correspondence problem. Valid and invalid computations of Turing machines, undecidable properties of context free language problems, Basics of Recursive function theory.

References

1. *C.Papadimitrou and C.L.Lewis "Elements of Theory of Computation", PHI*
2. *J.E.Hopcroft and J.D.Ullman "Introduction to Automata Theory, Languages of Computations", Addison-Wesley*

Course Objectives:

CO1	Analyzing problems, and designing and implementing algorithmic solutions
CO2	Solving problems properly, achieving an implementation that is correct, effective and efficient.
CO3	Using computers at user level, including operative systems and programming environments
CO4	Knowledge of computer equipment, including both hardware and software
CO5	Identifying information needs to solve problems, recovering information and applying it to the resolution